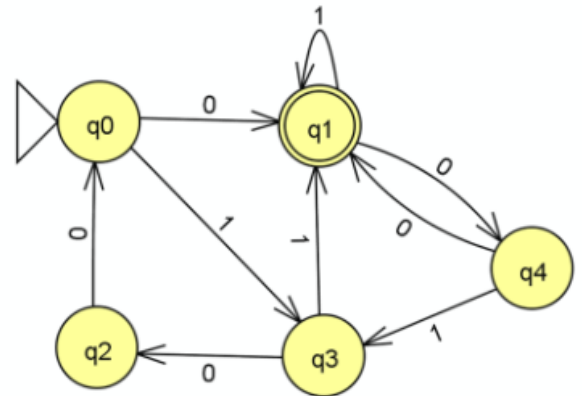


AA2 – DFAs

(soluções)

1



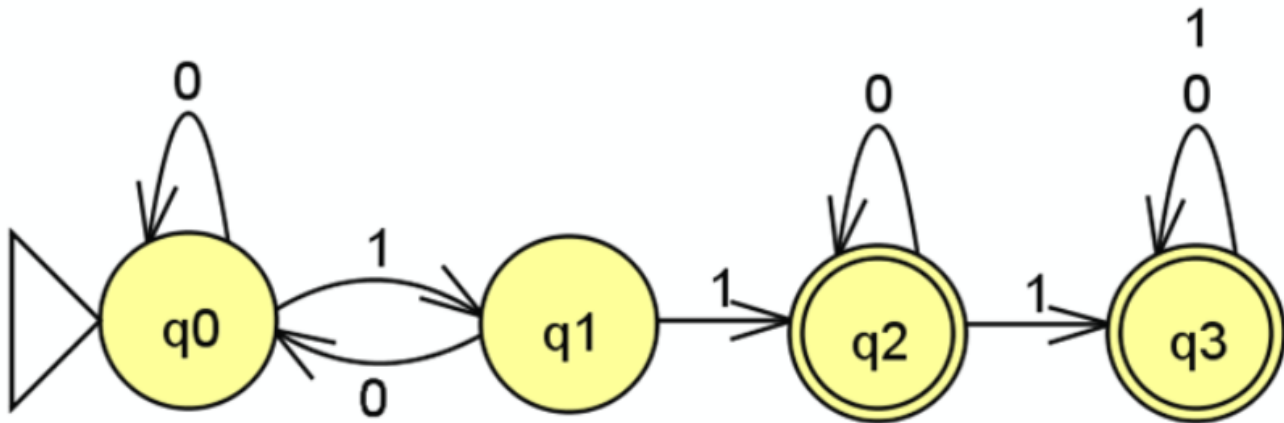
Consider the following automaton. Which of the following extended transition functions arrives at a dead state?

(1/1 Ponto)

- ☐ $\delta^*(q_0, 1101110)$
- ☒ $\delta^*(q_0, 0110101)$
- ☐ $\delta^*(q_0, 0100110)$
- ☐ $\delta^*(q_0, 1100001)$

2

Consider the following automaton. Which of the following options describes the language accepted by it? 
(1/1 Ponto)



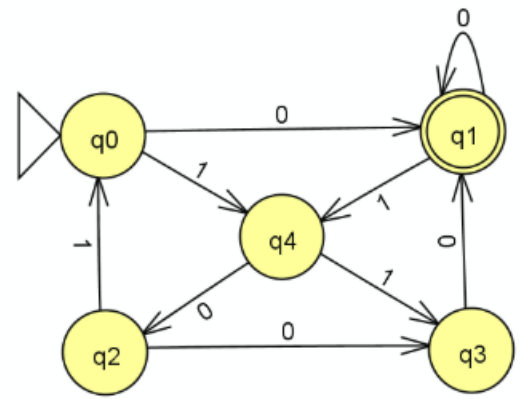
- ☐ Strings over the alphabet {0, 1} with 11 at the end of the string
- ☐ Strings over the alphabet {0, 1} with 111 at the end of the string
- ☒ Strings over the alphabet {0, 1} with 11 in any place of the string
- ☐ Strings over the alphabet {0, 1} with 111 in any place of the string

3

An automaton stops when it reaches an acceptance state. 
(1/1 Ponto)

- ☐ True
- ☒ False

4

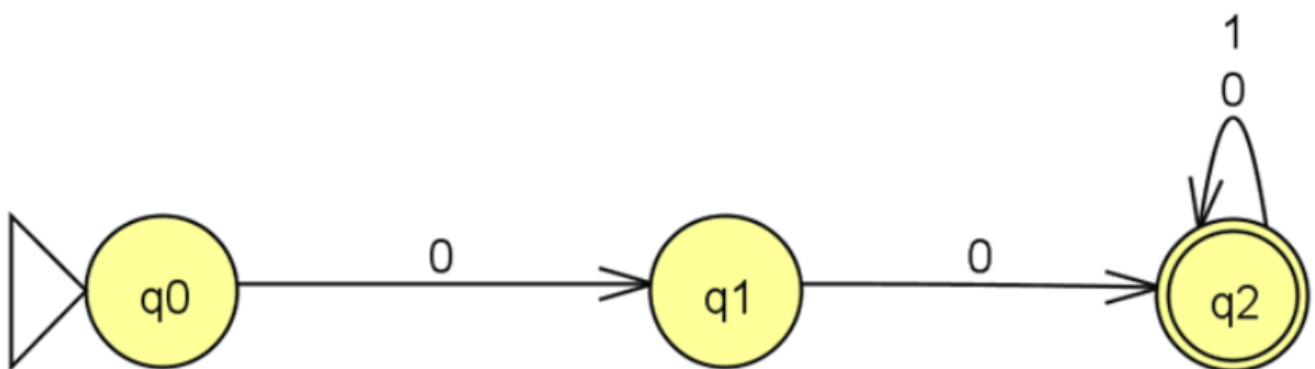


Consider the following automaton. What is the best classification for it?
(1/1 Ponto)

- ☐ Complete DFA
- ☒ Incomplete DFA
- ☐ None of the above

5

Consider the following automaton. Which of the following options describes the language accepted by it? 
(1/1 Ponto)



- ☒ Strings over the alphabet {0, 1} that start in 00.
- ☐ Strings over the alphabet {0, 1} that end in 00.
- ☐ Strings over the alphabet {0, 1} with 00 as substring.

6

Consider the DFA 1 with initial state A and acceptance state B, and DFA 2 with initial state C and acceptance D. After applying the cartesian product we obtain a new DFA with states {AC, AD, BC, BD}.

Which states should be the acceptance states if we want to obtain DFA 2 - DFA 1? 
(1/1 Ponto)

☐ BD, AD, BC

☒ AD

☐ BC

☐ BD

7

You can only represent finite languages with DFAs.
(1/1 Ponto)

☐ True

☒ False
